

Hospital Bed Allocation Architecture

Architectural Choice: We selected the Microservices Architecture.

Two Reasons:

1. **Fault Isolation:** By separating the application into independent operational units—such as the Triage Engine Component and the Bed Management Component—a failure or bottleneck in one service (like the Notification Component) will not disrupt critical hospital functions like triage scoring and patient bed assignments.
2. **Independent Scalability:** Distinct modules experience different workloads. The Notification Component handling the WebSocket API can be dynamically scaled independently of the Healthcare Database Component during mass casualty events or peak emergency hours to handle high concurrent connections without wasting resources on the entire system.

Security Advantage: Microservices naturally enforce strict boundaries between domains, allowing for granular security controls. By separating the Dashboard UI Component from backend services like the Triage Engine Component, Role-Based Access Control (RBAC) can be strictly enforced at the Triage API and Bed Assignment API levels. This ensures that sensitive Protected Health Information (PHI) is isolated and only accessible to authorized clinical staff making direct API calls.

Performance Benefit: Utilizing a dedicated Notification Component with a WebSocket API offloads the overhead of real-time broadcasting from core transactional services. This architectural separation ensures that high-frequency real-time dashboard updates are pushed instantly without being delayed by the intensive processing happening within the Triage Engine Component or heavy read/write locks in the Healthcare Database Component.